Assessment of the Profile-Cached Blocking Calculus for $\mathcal{ALCI}_{RCC5}$ under Strong EQ

Response to GPT's draft series (April 2, 2026)

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Disclaimer. This assessment was authored entirely by Claude (Anthropic), an AI assistant. The evaluation and the claims made herein have **not been verified by human domain experts**. Both the assessed drafts and this response are discussion pieces.

Abstract

GPT has produced a series of four draft notes proposing a profile-cached global blocking calculus for $\mathcal{ALCI}_{RCC5}$ under strong EQ semantics. The series exhibits an impressive self-correction cycle: an initial conditional result is refined through three revisions that correct a false lemma, provide an explicit finite signature system, and ultimately claim a full replay theorem based on a meet-semilattice structure on RCC5 witness labels.

We assess each note in turn. The **local combinatorial results are correct and novel**: the counterexample to flat normalization, the coherent-block normalization lemma, the meet-polymorphism property of the RCC5 composition table, and the robust colorwise normalization theorem all withstand scrutiny and computational verification. However, the **global unraveling argument in the final note has a genuine gap**: the unraveling changes the predecessor color structure (multiplicities, inter-color relations, and even the color alphabet itself), so the replay theorem—proved for nodes with equal signatures in the finite tableau—cannot be directly applied to the unraveling's copies.

We exhibit this gap concretely on the canonical infinite-model concept $C_\infty = \exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top$. We then discuss how the gap might be closed, noting that a patchwork-based argument for cross-chain edge assignment could complement GPT's local machinery.

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*Claude is an AI assistant by Anthropic. This assessment was prepared at the request of Michael Wessel (`miacwess@gmail.com`). It evaluates four draft notes by GPT proposing a profile-cached blocking calculus for $\mathcal{ALCI}_{RCC5}$ under strong EQ semantics.

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1 Introduction

The decidability of concept satisfiability in $\mathcal{ALCI}_{RCC5}$ under strong EQ semantics has been an open problem since Wessel's report [7]. Multiple approaches have been attempted:

- A quasimodel method with type elimination [1], which has the *extension gap* in the soundness direction (open quasimodel need not yield a model);
- A contextual tableau [4], whose completeness conjecture $\mathrm{FW}(C, N)$ is false [3];
- A direct model construction from open completion graphs [2], which identifies the root cause as the self-absorption failure $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ and proposes a resolution (with caveats on the path-consistency argument).

GPT now proposes a fourth approach: a **profile-cached global blocking calculus** in the style of anywhere blocking / global caching, adapted to complete-graph semantics. The approach is developed across four draft notes:

1. **Blocking Draft** — initial proposal, conditional on a classwise witness-normalization lemma;
2. **Blocking Revised** — self-correction: flat normalization is false; repaired via coherent predecessor blocks;
3. **Explicit Signatures** — concrete depth-indexed signature system with finite-index proof;
4. **Replay Final** — claims full replay theorem via meet-semilattice normalization, yielding decidability.

This note assesses the four papers in order, verifies the local algebraic claims computationally, and identifies the remaining gap.

2 Paper 1: The Initial Draft

2.1 Architecture

The draft proposes a global blocking scheme for complete-graph tableaux. Each saturated node x receives a *contextual profile*

$$\mathsf{Prof}(x) = \big(\lambda(x),\ \mathsf{Ctx}(x)\big),$$

where $\lambda(x)$ is the saturated Hintikka type and $\mathsf{Ctx}(x)$ is a finite quotient of the older neighborhood, collapsing nodes with identical *context signatures* $(\lambda(u), \rho(u,x), \Gamma_{\rho(u,x)}(\lambda(u)))$.

A node y is blocked by an older node x when $\mathsf{Prof}(y) = \mathsf{Prof}(x)$. Existential witnesses are recorded as *witness recipes*: a target type plus a classwise assignment of RCC5 labels from each context class to the new witness. Termination follows from the finiteness of the profile universe.

2.2 Assessment

The architecture is sound and well-motivated. The observation that blocking must be global (not ancestor-based) is correct: in complete-graph models, every older node constrains every newer node, so tree-based blocking is inapplicable.

The paper is honest about its status: all results are conditional on a **classwise witness-normalization lemma** (Lemma 4.1 in the draft), which states that witnesses can be chosen so that their behavior depends only on context classes, not individual representatives. This is correctly identified as the key proof obligation.

Verdict: clean conditional result. The metatheory is correct modulo the stated assumption.

3 Paper 2: The Self-Correction

3.1 The counterexample to flat normalization

The most important contribution of Paper 2 is Proposition 4.2: flat classwise normalization is **false**.

Example 3.1 (GPT's counterexample, verified)**.** Node x carries $\exists\mathsf{DR}.(p \sqcap q \sqcap r)$. Older nodes a_1, a_2, b_1, b_2, c are all DR-related to x. Types:

$$\tau_A : \forall\mathsf{PO}.\neg p,\ \forall\mathsf{PP}.\neg p,\ \forall\mathsf{PPI}.\neg p; \quad \tau_B : \forall\mathsf{PO}.\neg q,\ \forall\mathsf{PP}.\neg q; \quad \tau_C : \forall\mathsf{DR}.\neg r,\ \forall\mathsf{PP}.\neg r,\ \forall\mathsf{PPI}.\neg r.$$

All inter-predecessor edges are DR except $b_2\ \mathsf{PPI}\ c$. Target type $t = \{p, q, r, \top\}$, witness labels $\eta(a_i) = \mathsf{DR}$, $\eta(b_1) = \mathsf{DR}$, $\eta(b_2) = \mathsf{PPI}$, $\eta(c) = \mathsf{PO}$.

Verification. The payload constraints force:

- From τ_A: only DR is allowed (all other payloads contain $\neg p$, but $p \in t$).
- From τ_B: only DR or PPI are allowed (PO and PP payloads contain $\neg q$).
- From τ_C: only PO is allowed (DR, PP, PPI payloads contain $\neg r$).

The triangle constraint for b_1 (with c): $\mathsf{Tri}(\mathsf{DR}, \mathsf{PO}, \eta(b_1))$ requires $\eta(b_1) \in \mathsf{comp}(\mathsf{DR}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$. Combined with the payload restriction to $\{\mathsf{DR}, \mathsf{PPI}\}$, we get $\eta(b_1) = \mathsf{DR}$ (forced).

The triangle constraint for b_2 (with c): $\mathsf{Tri}(\mathsf{PPI}, \mathsf{PO}, \eta(b_2))$ requires $\eta(b_2) \in \mathsf{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PO}, \mathsf{PPI}\}$. Combined with the payload restriction to $\{\mathsf{DR}, \mathsf{PPI}\}$, we get $\eta(b_2) = \mathsf{PPI}$ (forced).

Since b_1 and b_2 share the same flat class but are forced to different labels, no flat-classwise normalization exists. $\square$

Remark 3.2 (Connection to our self-absorption analysis)**.** The root cause is precisely the self-absorption failure we identified independently: $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$. The edge $b_2\ \mathsf{PPI}\ c$ (versus $b_1\ \mathsf{DR}\ c$) changes the triangle constraint, forcing different witness labels despite identical flat classes. GPT's counterexample and our algebraic analysis converge on the same structural rigidity in the RCC5 composition table.

3.2 Coherent-block normalization

Paper 2 introduces **coherent predecessor blocks**: a refinement of flat classes where inter-block relations are uniform and intra-block relations are uniform symmetric (DR or PO).

Proposition 3.3 (Coherent-block normalization — verified)**.** *Every locally admissible symbolic extension (t, η) can be normalized to (t, η') that is constant on each coherent block, using only labels already realized in the original extension.*

The proof proceeds by cases on the intra-block label:

- **PO-blocks**: Every label r is self-stable under PO: $\mathsf{Tri}(\mathsf{PO}, r, r)$ holds for all $r \in \mathcal{R}$. (Verified: $r \in \mathsf{comp}(\mathsf{PO}, r)$ for all r.) Any realized label works.
- **DR-blocks**: PPI is the only unstable label: $\mathsf{Tri}(\mathsf{DR}, \mathsf{PPI}, \mathsf{PPI})$ fails because $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$. So in a non-singleton DR-block, not all members can carry PPI (two PPI-labeled members would violate the triangle condition). Therefore some label in $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ is realized, and all three are DR-stable.
- **Inter-block**: By coherence (uniform edge q_{BC}), one can pick representatives $u \in B$, $v \in C$ realizing the chosen labels and appeal to the original admissibility.

Verdict: correct and clean. This is the right local repair.

4 Paper 3: Explicit Signatures and Finite Index

Paper 3 (the "explicit signatures" addendum) makes the depth-indexed signature construction concrete and proves the finite-index lemma. The construction uses:

- Rank-0 signatures = propositional types $\mathsf{tp}_0(x)$.
- Rank-$(k{+}1)$ predecessor colors built from rank-k signatures, incoming relation, truncated payload, refined through $k{+}1$ rounds of relation-set profiling.
- Context tables: multiplicity map μ (values in $\{0, 1, +\}$) and relation-set table Q.
- Projected witness footprints recording which labels occur on coherent blocks of each final color.

The finite-index lemma is correct: at each rank, all components range over finite domains, so the signature space is finite.

The paper is deliberately cautious: it proves finiteness of the signature system but does **not** claim the replay theorem. It explicitly states that turning projected footprints into functional replay recipes is the remaining step.

Verdict: correct. The finite-index part of the blocking program is settled.

5 Paper 4: The Meet-Semilattice and Replay Theorem

Paper 4 claims to close the last gap. The key innovations are:

1. Predecessor colors now include the **full payload vector** $(\Gamma_R(\lambda(u)) \cap \mathsf{cl}_k(C))_{R \in \mathcal{R}}$, not just the incoming-edge payload.
2. Witness footprints are sharpened to **robust colorwise recipes** $\chi : A_{k+1} \to \mathcal{R}$ using a meet-normalization.

5.1 The meet-semilattice structure (verified)

The four non-EQ RCC5 labels are partially ordered by the diamond:

$$\mathsf{DR} < \mathsf{PO} < \mathsf{PP}, \qquad \mathsf{DR} < \mathsf{PO} < \mathsf{PPI}, \qquad \mathsf{PP} \text{ and } \mathsf{PPI} \text{ incomparable.}$$

The meet (greatest lower bound) gives:

$$\mathsf{DR} \wedge r = \mathsf{DR}, \quad \mathsf{PO} \wedge \mathsf{PP} = \mathsf{PO}, \quad \mathsf{PO} \wedge \mathsf{PPI} = \mathsf{PO}, \quad \mathsf{PP} \wedge \mathsf{PPI} = \mathsf{PO}.$$

Lemma 5.1 (Meet polymorphism — computationally verified)**.** *For every $q \in \mathcal{R}$, the relation $R_q = \{(r, s) \mid \mathsf{Tri}(q, s, r)\}$ is closed under componentwise meet.*

We verified all $4 \times 16 \times 16 = 1{,}024$ cases computationally against the full RCC5 composition table under strong EQ. Zero violations.

Lemma 5.2 (No PP/PPI within one final color — correct)**.** *If two predecessors of the same final color were related by* PP*, every node of that color would need an outgoing* PP*-edge within the color class. But* PP *on a finite set is a strict partial order, which has maximal elements with no outgoing* PP*-edge. Contradiction.*

Lemma 5.3 (Realized labels cannot mix PP and PPI — correct)**.** *If coherent blocks of the same color carried both* PP *and* PPI *as witness labels, the triangle condition would require* $\mathsf{Tri}(q, \mathsf{PPI}, \mathsf{PP})$ *for* $q \in \{\mathsf{DR}, \mathsf{PO}\}$ *(the only possible intra-color relations by the previous lemma). Verified:* $\mathsf{PP} \notin \mathsf{comp}(q, \mathsf{PPI})$ *for both* $q = \mathsf{DR}$ *and* $q = \mathsf{PO}$*. Contradiction.*

Theorem 5.4 (Robust colorwise normalization — correct)**.** *For each final color* a*, the meet* $m(a) = \bigwedge\{\eta(B) \mid B$ *coherent block of color* $a\}$ *is in the realized set (by the previous lemma) and yields a locally admissible extension. The proof uses the covering argument: for each pair of colors* a, b *and relation* $q \in Q(a, b)$*, choose a covering family of block pairs, apply admissibility to each, and take the componentwise meet (which stays in* R_q *by meet polymorphism).*

Verdict on local results: correct, elegant, and novel. The meet-semilattice structure is a genuinely nice algebraic insight that we had not identified in our own analysis.

5.2 A missing admissibility constraint (minor)

The definition of *locally admissible symbolic extension* in all four papers checks only:

(i) Payload preservation: $\Gamma_{\eta(u)}(\lambda(u)) \subseteq t$ for $u \in \mathsf{old}(x)$.

(ii) Pairwise triangles: $\mathsf{Tri}(\rho(u, v), \eta(v), \eta(u))$ for distinct $u, v \in \mathsf{old}(x)$.

However, when a witness y is created for $\exists S.E$ at node x, there is also the triangle (u, x, y) for each $u \in \mathsf{old}(x)$:

$$\eta(u) \in \mathsf{comp}(\rho(u, x),\ S). \tag{1}$$

This constraint is not listed in the admissibility definition.

This is a **presentation issue, not a correctness issue**. We verified computationally that for every $q, S \in \mathcal{R}$, the set $\mathsf{comp}(q, S)$ is closed under the diamond meet. Therefore the meet-normalization preserves this constraint even though it is not formally tracked. Moreover, model-extracted recipes automatically satisfy (1). The fix is simply to add this constraint to the definition.

6 The Unraveling Gap

The critical issue is Theorem 6.2 of Paper 4 (the "Blocked Unraveling Theorem"), which claims that an open saturated blocked branch can be unraveled into a genuine model. The proof is a one-paragraph sketch asserting that "standard induction on formula depth" suffices.

The replay theorem (Theorem 6.1) is correct: if two nodes in the finite tableau have equal rank-d signatures, then robust recipes at one can be applied at the other. The gap is that **nodes in the unraveling need not have the same signature as the representative they are copies of**.

6.1 Concrete demonstration: C_∞

Consider $C_\infty = \exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$, which requires an infinite model (every element must have a PP-successor, and so must every PP-descendant).

The finite tableau.

- Node r_0: type $\tau = \{\exists\mathsf{PP}.\top,\ \forall\mathsf{PP}.\exists\mathsf{PP}.\top\}$, no predecessors ($\mathsf{old}(r_0) = \emptyset$).
- Node r_1: PP-witness for r_0, type τ (forced by the $\forall\mathsf{PP}$-propagation), predecessor set $\mathsf{old}(r_1) = \{r_0\}$.
- r_1 is blocked by r_0 (same profile in a suitable sense).

The recipe at r_1 for the existential $\exists\mathsf{PP}.\top$ assigns a label to its sole predecessor r_0. The triangle $(r_0, r_1, \text{witness})$ with $\rho(r_0, r_1) = \mathsf{PP}$ and $S = \mathsf{PP}$ forces $\eta(r_0) \in \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, so $\eta(r_0) = \mathsf{PP}$.

The unraveling. The unraveling creates an infinite PP-chain:

$$c_0 \xrightarrow{\mathsf{PP}} c_1 \xrightarrow{\mathsf{PP}} c_2 \xrightarrow{\mathsf{PP}} c_3 \xrightarrow{\mathsf{PP}} \cdots$$

At step 2, c_2 is created as a PP-witness for c_1. The recipe assigns $\rho(c_0, c_2) = \mathsf{PP}$ (from the recipe at r_1).

At step 3, c_3 is created for c_2. Now $\mathsf{old}(c_2) = \{c_0, c_1\}$. Both predecessors have initial color $(\tau, \mathsf{PP}, \mathsf{Pay}(\tau))$ relative to c_2 (since $\rho(c_0, c_2) = \mathsf{PP}$ and $\rho(c_1, c_2) = \mathsf{PP}$). After one refinement round:

- c_0 sees c_1 (same initial color) via PP, so its relation-set profile includes PP to its own initial color.
- c_1 sees c_0 (same initial color) via PPI, so its relation-set profile includes PPI to the same initial color.

Since $\mathsf{PP} \neq \mathsf{PPI}$, the two predecessors get **different final colors** after one refinement round.

The mismatch. In the finite tableau, the representative r_1 had *one* predecessor (r_0) of one color, with multiplicity 1. In the unraveling, c_2 has *two* predecessors of *two different* colors, each with multiplicity 1. The multiplicity maps differ; the relation-set tables differ (the unraveling has a PP/PPI inter-color relation that did not exist in the finite tableau); and the recipe at r_1 has no assignment for the new color because that color did not exist in r_1's context.

At step n, the situation is worse: c_n has n predecessors, which after color refinement split into $O(n)$ distinct final colors. The recipe, designed for a bounded neighborhood, cannot cover them.

Conclusion. The replay theorem applies to nodes with *equal signatures*. But copies in the unraveling accumulate predecessors whose color structure diverges from the finite tableau's. The replay theorem's prerequisites are not met, and the unraveling construction does not produce a certified model.

6.2 Why this gap is non-trivial

The gap is structural, not cosmetic. In tree-based DL tableaux, unraveling is straightforward because each node's neighborhood is its parent and children—a bounded, predictable structure. In complete-graph tableaux for $\mathcal{ALCI}_{RCC5}$, every node is connected to every other node. The unraveling must assign RCC5 labels to *all pairs* of copies, including cross-chain pairs whose existence was not anticipated by the finite tableau's recipes.

This is the same fundamental difficulty that affects all three previous approaches:

- In the quasimodel approach, it manifests as the extension gap (abstract quasimodels need not yield models).
- In the contextual tableau, it manifests as the failure of $\mathrm{FW}(C, N)$ (local states of bounded width cannot capture the global model).
- In our direct model construction, it manifests as the path-consistency argument for the cross-subtree constraint network (Cases B and C).

GPT's profile-cached approach elegantly handles the *local* problem (what labels to assign from predecessors to a new witness) but does not fully address the *global* problem (what labels to assign between copies that were not simultaneously present in the finite tableau).

7 Possible Repairs

The gap is specific and the local machinery is strong. Several repair strategies are conceivable.

7.1 Patchwork-based cross-chain assignment

Our companion paper [2] handles cross-subtree edges by setting up a *disjunctive RCC5 constraint network* over all pairs of copies and appealing to **full RCC5 tractability** (Renz & Nebel [6]): a path-consistent disjunctive RCC5 network is globally consistent.

A hybrid approach could work as follows:

1. Use GPT's profile-cached blocking for **termination** and **local recipe generation**.
2. When unraveling, assign **tree edges** (parent-to-child via recipes) as in GPT's construction.
3. For **cross-chain edges** (between copies not in a parent-child relationship), set up a disjunctive constraint network whose constraints come from the composition table applied along the tree path connecting the two copies.
4. Prove **path-consistency** of this network using the tableau's properties and the meet-normalization results.
5. Apply full RCC5 tractability to obtain a globally consistent atomic refinement.

This would marry GPT's local elegance with the global consistency guarantee from the patchwork property.

7.2 Weakened replay with monotone signatures

Alternatively, one could define a notion of *signature subsumption* such that a copy c in the unraveling has a signature that *subsumes* its representative's signature (more colors, richer context). If one can show that recipes designed for the simpler signature remain admissible under the richer one—using the monotonicity of the meet-normalization and the $\mathsf{comp}(q, S)$-meet-closure we verified—then the replay theorem could be extended to subsumption rather than equality.

This seems plausible but would require careful formalization of the subsumption relation and its interaction with the recipe admissibility conditions.

7.3 Truncated unraveling with depth-bounded predecessors

A third option: instead of unraveling the full cache, build a *truncated* unraveling where each copy only remembers predecessors up to some bounded depth in the unraveling tree. The truncation depth would be chosen so that the signature stays within the finite alphabet. Cross-chain edges beyond the truncation horizon would be assigned by a default rule (e.g., DR, which is always composition-consistent with itself). The correctness of the truth lemma under truncation would need to be verified.

8 Summary of Verification Results

Claim	Paper	Status
Flat normalization is false (counterexample)	2	**Correct**
Coherent-block normalization	2	**Correct**
Explicit finite-index signature system	3	**Correct**
Meet polymorphism (R_q closed under $\wedge$)	4	**Verified** (1,024 cases)
No PP/PPI within one final color	4	**Correct**
Realized labels don't mix PP and PPI	4	**Correct**
Robust colorwise normalization	4	**Correct**
$\mathsf{comp}(q, S)$ is meet-closed (missing constraint)	—	**Verified** (64 cases)
Replay theorem (finite tableau)	4	**Correct**
Blocked unraveling theorem	4	**Gap**

9 Comparison of Approaches

	GPT: profile-cached blocking	**Claude: direct model construction**
Local arguments	Meet normalization, coherent blocks — **proved**	Self-absorption, forced DR edges — **proved**
Global model construction	Unraveling from finite cache — **gap**	Tree unraveling + constraint network — **gap** (Cases B/C)
Key algebraic insight	Meet-semilattice on RCC5 labels	Self-absorption failure $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$
Use of patchwork / tractability	Not used	Central (full RCC5 tractability)
Blocking / termination	Explicit (profile-cached, proved)	Inherited from original tableau

The two approaches have **complementary strengths**. GPT's meet normalization elegantly solves the local witness-recipe problem; our patchwork-based argument addresses the global edge-consistency problem. A synthesis using GPT's blocking for termination and patchwork for cross-chain edges in the unraveling could potentially close both remaining gaps.

10 Conclusion

GPT's draft series represents strong and largely correct work on a genuinely difficult problem. The self-correction cycle—from a false lemma through coherent blocks to the meet-semilattice—is a model of honest, iterative proof development.

The local results (coherent-block normalization, meet polymorphism, robust colorwise normalization, finite-index signatures, replay theorem for the finite tableau) are all correct, verified, and novel. The meet-semilattice structure on RCC5 witness labels is an insight that neither we nor GPT's earlier contextual tableau had identified.

The global unraveling argument, however, has a genuine gap. The unraveling changes the predecessor color structure in ways that prevent direct application of the replay theorem. This

is the same structural difficulty—in different guise—that affects all approaches to $\mathcal{ALCI}_{RCC5}$ decidability.

We believe the gap is closable, most likely by incorporating a patchwork-based argument for cross-chain edge assignment. The combination of GPT's profile-cached blocking (for termination and local recipes) with the patchwork property (for global consistency) appears to be the most promising path toward a complete proof.

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